

Predicting the Next Category and Region of a Visit: A Check-in-Level Multi-Task Study on Mobility Data

Vitor H. O. Silva, Germano B. dos Santos, Fabrício A. Silva
Laboratório de Inteligência em Sistemas Pervasivos e Distribuídos (NESPED-LAB)
Universidade Federal de Viçosa, Florestal, MG, Brazil
E-mail: {vitor.h.oliveira, germano.santos, fabricio.asilva}@ufv.br

Abstract—Location-based social networks (LBSNs) record where people go and what they do, one check-in at a time. If we can anticipate the next visit, mobile and urban services can act ahead of demand. Two coarse questions about the next visit are usually enough: its category (the kind of place) and its region (which part of the city). These are normally handled by separate models. We therefore ask whether one model can learn both, and what sharing a single model costs. We first build a check-in-level representation that describes each visit in its own context, instead of giving every place one fixed vector. Across six datasets, five U.S. states and one non-U.S. city (Istanbul), this lifts next-category prediction by a wide margin over a standard place embedding (about +28 to +40 macro-averaged F1), and most of the gain comes from the per-visit context. We then train one multi-task model for both tasks. At every dataset it outperforms a dedicated category model (about +5 to +9 macro-F1), and on region it outperforms the dedicated region model at four of the six datasets, while matching it (statistically, within two points) at the other two. Across the five U.S. states the gain on region grows with the number of regions, with the two largest states measured at a single seed. On the non-U.S. city the result holds: the joint model outperforms on category and comes out slightly ahead on region.

Index Terms—next-category prediction, next-region prediction, multi-task learning, check-in-level representation, location-based social networks, mobility data

I. INTRODUCTION

Location-based social networks, such as Gowalla [1], let people share the places they visit as check-ins, which together record how people move through a city. Individual mobility is highly predictable in principle [2], so a service that anticipates the next move can prepare ahead of time, staging content where a user is heading [3] or planning capacity there before demand arrives. Predicted handovers already let cellular services adapt in advance at the network level [4]; at the city scale, check-in mobility analysis is likewise a design input for mobility-aware services [5].

Predicting the next place exactly, a single point of interest (POI), is hard and often more than a service needs. Two coarser questions are usually enough: the next category, the kind of place such as food or shopping, and the next region, the part of the city a user is heading to (a neighborhood-scale unit such as the U.S. census tract). One captures intent, the other geography; we study whether one model should learn both at once.

Sharing one representation across tasks is not free: in multi-task learning (MTL), one model does several jobs at once by sharing most of its parts, so the shared parameters can settle on a compromise that suits neither task perfectly, helping one task while hurting the other [6]. Our earlier work [7] observed exactly this for next-category and next-region; the useful question is where sharing helps, where it costs, and how to share so the gains hold and the cost stays small.

We introduce two changes. First, we build a check-in-level representation: instead of one fixed vector per place, each check-in gets its own vector carrying its context (the time, nearby places, and recent trail). This adds a fourth, check-in level beneath the place, region, and city levels of hierarchical graph infomax [8], [9] (Fig. 1). Per-visit context is not new on its own; what is new is obtaining it from inside this hierarchical graph representation, a choice we test directly (Section VI-A). Second, we train one model that predicts the next category and the next region in a single forward pass, with a shared trunk (a cross-attention stack where the two tasks exchange semantic context) and a private spatial path for the region task (Fig. 2). We evaluate on two deliberately different settings: five U.S. states of different sizes (Gowalla) and one international city (Istanbul).

Our contributions are the following.

- A check-in-level representation that extends hierarchical graph infomax from the place to the individual visit, improving next-category prediction over a standard place embedding by about +28 to +40 points of macro-F1 (a balanced score that counts every category equally) across all six datasets (Table II). A controlled test attributes most of the gain to the per-visit context, not to extra training signal. The margin is large because a single fixed vector per place cannot separate places that host several kinds of activity, which per-visit context recovers.
- A single model for joint next-category and next-region prediction, sharing semantic context across tasks while keeping a private path for the spatial one. In one forward pass it outperforms a dedicated single-task category model at every dataset (by about +5 to +9 points of macro-F1), even though each dedicated model is tuned per dataset. On region, it outperforms at four of the six datasets and stays statistically non-inferior within a

two-point margin (TOST, a test of equivalence) at the remaining two (Table III).

- An empirical account, across five states and one international city, of how the joint gain scales with the number of regions. The category gain holds everywhere, and across the U.S. states the region result moves monotonically, from a non-inferior match (TOST, ± 2 percentage points, or pp) at the small region counts to outperforming the dedicated model at the large, with the largest state (California) showing the largest region gain (Fig. 3); Istanbul, the smallest dataset, also comes out ahead on region. Four of the six datasets are measured at four seeds over five folds on both arms (the joint model and the dedicated one); California and Texas are single-seed and provisional.

The rest of the paper presents related work (Section II), the problem (Section III), the method and setup (Sections IV and V), results (Section VI), and discussion and conclusions (Sections VII and VIII).

II. BACKGROUND AND RELATED WORK

A. From place embeddings to per-visit embeddings

A place embedding turns a POI into a vector so that similar or nearby places sit close together, rather than each being a bare, arbitrary index. Deep Graph Infomax (DGI) [9] learns such vectors self-supervised, training them to tell the real place network, linked by similarity, time, and distance, from a shuffled copy. Hierarchical Graph Infomax (HGI) [8] adds geographic structure, organizing the network as place, then region, then city. Both produce one vector per place, so two visits to the same coffee shop look identical to the model. Contextual check-in representations instead give each visit its own vector: CTLE [10], the closest example, learns a vector per visit by masking and reconstructing parts of a user’s check-in sequence. Our representation, Check2HGI, also works at the check-in level but gets there differently: it keeps the hierarchical place-to-region-to-city network and the same infomax objective, now extended one level deeper, from individual places down to individual check-ins (Fig. 1), rather than switching to a sequence model as CTLE does.

The novelty is this specific combination: per-visit context inside a hierarchical graph-infomax representation. We compare against CTLE directly to show that the combination, not per-visit context on its own, is the source of the gain.

B. Predicting the next category and the next region

A large body of work predicts the next place a user will visit, the exact POI, with recurrent and attention models such as DeepMove [11], STAN [12], and GETNext [13]. In contrast, we target two properties of the next visit, its category and its region, rather than the exact place. We choose them because they are what a mobility-aware service can act on, not to make the task easier; the region alone spans hundreds to several thousand classes. Predicting over a partition of the map is also the standard formulation in the human-mobility literature, with a grid cell as the target [14]; our next-region

task substitutes administrative units for grid cells. The category task itself follows our earlier line of work [7]. The field increasingly models several granularities at once; in that work, category and region are auxiliary signals that help a headline next-place task (MCMG [15], HMT-GRN [16]). We study the pair as the object itself. To our knowledge, fine-grained region as a co-equal end target, rather than an auxiliary coarse cell, is underexplored. The nearest exceptions stop short of our pairing: DRRGNN [17] forecasts a person’s next activity region jointly with a mobility-intention label, but over regions discovered per person rather than a fixed citywide partition; a generative recommender [18] predicts category and region jointly, but only as auxiliary steps toward its next-place ranking.

C. Multi-task learning for mobility

MCARNN [19] jointly predicts activity and location, and a recent hierarchy-aware model continues the pattern [20]; in both the location target is the exact place. The closest alternative to our setting is the cascade: predict the category first, then the place given the category [21]. CSLSL [22] is its current form, predicting in a chain (when, then what, then where), with location as the headline output and category as an instrumental step; CatDM [23] likewise uses category instrumentally, to filter candidate places. We instead predict category and region in parallel, as co-equal end targets in one model with a single forward pass, and we drop the next-place target entirely, so neither task is instrumental to a third; a chain would have to promote one co-equal target to an upstream stage whose errors the other inherits, an order nothing in the tasks dictates. Since the cascade is the pattern the field actually uses, we test the choice directly (Section VI-B). On optimization we are deliberately conservative: joint training with a fixed loss weighting is standard practice [6], not itself our contribution. Reports find that gradient-balancing methods, which re-weight the tasks’ updates on the fly (PCGrad [24], Nash-MTL [25]), rarely improve on a well-tuned fixed weighting at two tasks [26]. We confirm this: none of the balancers we tried, including the two named above, improved on a tuned fixed task weighting in our model. The reason is visible in the gradients: the next-category and next-region updates on the shared layers are near zero-correlated throughout training, so a balancer has no conflict to resolve, consistent with the mechanism test in Section VI-B; this is a finding for this pair of tasks, not a general rule.

III. PROBLEM AND TASKS

We order each user’s check-ins in time, form windows of nine consecutive visits, and from each window predict two properties of the next visit. The first is its *category*, one of seven fixed labels: Community, Entertainment, Food, Nightlife, Outdoors, Shopping, and Travel. Istanbul’s source collection maps its places onto the same seven labels (Section V-A). The second is its *region*, the place’s census tract (the *mahalle* for Istanbul). We do not predict the exact next place; these are easier to learn and, for most uses, enough. Region,

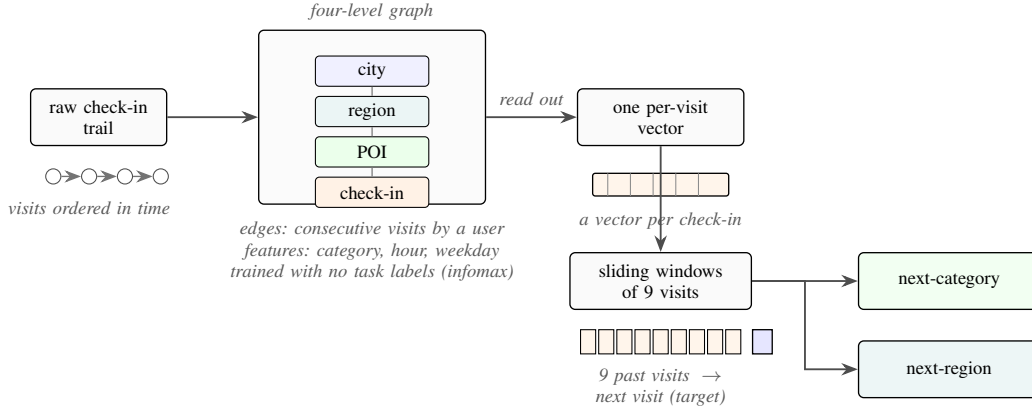


Fig. 1. From a check-in trail to two predictions: each visit is embedded in a four-level graph (check-in, place, region, city) and described in its own context; overlapping nine-visit windows feed a single model that outputs the next category and the next region.

unlike category, is a large label set: we frame next-region as classification over candidate census tracts, from about five hundred classes (Istanbul) to about eight thousand five hundred (California).

The motivation is practical: the category says what kind of place comes next, hence what a user will want; the region says where, hence where to prepare. The mobility literature supports such preparation, having long built city services on location-based social network traces [27]. A recent analysis of tourist check-in mobility finds that visits concentrate on a few hub places, argues that crowds and demand therefore concentrate, and ties this structure to infrastructure planning and adaptive strategies [5]. That analysis reads past visits; we predict two properties of the next one, and aggregated over users those predictions point to where demand is heading before it lands. A census tract is a neighborhood, not a radio cell, so we scope our claims to neighborhood-level preparation: demand and load anticipation, content staging, and capacity planning; not cell association or handover. We build and evaluate no such service here; Section VII quantifies what these predictions would give such a service.

IV. METHOD

A. The check-in-level representation

We want each visit, not each place, to carry its own vector. Figure 1 shows the full data flow, from check-ins to the two outputs.

We build a graph with four levels: the check-in, its place, the place’s region (a census tract), and the city. Edges tie the levels together and link a user’s consecutive check-ins, weighted by closeness in time; same-place visits connect through their shared place node one level up, and nearby places are linked at the place level. Each visit’s category, time of day, and day of week enter as input features of its node, not as edges. This keeps a hierarchical graph’s place-to-region-to-city geography and adds the check-in as a bottom level. We train the graph mainly with an infomax objective, under which each vector learns to match its real neighborhood and reject a shuffled

one [8], [9], plus two small label-free auxiliary terms: a masked reconstruction of each place’s aggregated category features and an anchor to a place embedding pre-trained, label-free, on the same data (weights 0.3 and 0.1). The training uses no task label: it never sees the next category or the next region. From the trained graph we read out one 64-dimensional vector per check-in (its region nodes carry trained vectors as well), so a model sees a sequence of per-visit vectors rather than repeated per-place ones.

Each ingredient appears in prior work on its own; as Section II-A argued, the contribution is their combination, and we show later that the combination, not extra supervision, makes the category task far easier to learn.

B. The single multi-task model

We then train one model that reads a window of recent per-visit vectors and predicts the next visit’s category and region at once. Figure 2 shows the design. Two input streams, a semantic stream with the window of per-visit vectors and a spatial stream in which each visit is represented by the trained vector of its region node from the same graph, pass through private per-task encoders (a small input network dedicated to each task, with no weights shared between them) into a shared cross-attention stack of two blocks, where the two streams exchange information: in each block, attention lets each stream read the other’s features, while each stream keeps its own feed-forward weights. The tasks therefore share by exchanging information between per-task streams, not by owning hidden layers in common. The stack then feeds a category output and a region output; the region output keeps a private spatial path, a small branch inside the one model (not a second model) that the category task does not touch.

The training loss is a fixed-weight sum of the two outputs, the category output weighted 0.75 and the region output 0.25, and both outputs use plain unweighted cross-entropy. Class-weighting, tested on both outputs, lowered both region accuracy and category macro-F1. This is ordinary fixed-weight joint training [6], kept deliberately plain so that any improve-

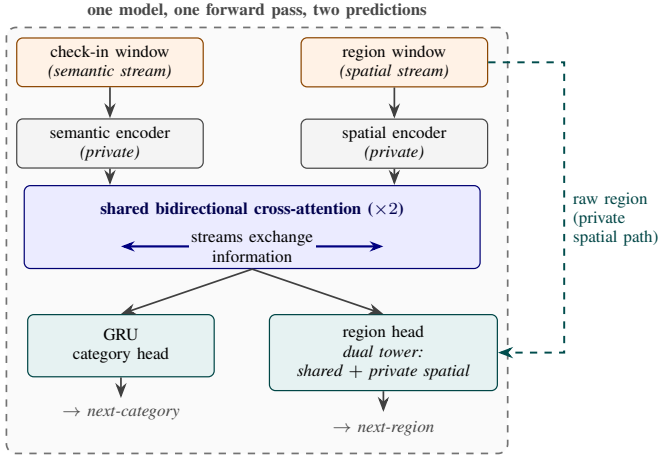


Fig. 2. The single multi-task model. A check-in (semantic) window and a region (spatial) window pass through private encoders into a shared cross-attention stack, the only place the two tasks interact; the category head predicts the next category, and the region head combines the shared output with a private spatial path. One model, one forward pass, two predictions.

ment over the dedicated single-task models comes from the shared representation, not from an adaptive weighting scheme.

The cost of serving both tasks this way is not free, and we state it plainly. The joint model carries the cross-attention stack and both task heads, so it is larger than either dedicated single-task model of Table III, and a forward pass costs more compute than running the two small dedicated models: about 4.2 million parameters at Alabama against 1.1 million for the two combined (5.2 against 2.0 at California). What the single model buys is operational rather than arithmetic: one artifact to train, version, and deploy, and one forward pass whose one set of inputs produces both answers at once.

V. EXPERIMENTAL SETUP

A. Data

Table I summarizes our six datasets. Five come from Gowalla, a location-based social network (collected 2009 to 2011), covering the U.S. states of Alabama, Arizona, Florida, Texas, and California [1]; the sixth is Istanbul, from the Massive-STEPS collection [28], a non-U.S. check. The states range from about 114 thousand check-ins and 1,109 regions in Alabama to several million check-ins and 8,501 regions in California. Every dataset uses the same seven place categories: Community, Entertainment, Food, Nightlife, Outdoors, Shopping, and Travel. The region unit is a census tract for the Gowalla states and a mahalle (a municipal neighborhood) for Istanbul.

B. Windows, splitting, and the integrity of the representation

Windows. For each user with at least ten visits we form time-ordered overlapping nine-visit windows, one starting at each visit, with the next visit as the target, giving both tasks more examples. Because many of a user’s windows would end on the same final visit, we keep one full-context window ending there and drop the padded duplicates.

Splitting. We split by user with stratified five-fold cross-validation, so all of a user’s windows fall in the same fold and overlap cannot leak: a test user’s visits never appear in training. The split is stratified on the next-category label because the seven-class task is imbalanced: Food is the majority class in every dataset, from about 25 percent of visits in Florida to 34 percent in Alabama.

Integrity of the representation. We train the representation once on the whole dataset and feed it to both the dedicated single-task models and the joint model; we argue, and verify, that it carries no usable information about the test visits, on three grounds. First, its objective is label-free, contrasting real graph neighborhoods against shuffled ones with small auxiliary terms reconstructing input-derived quantities; it never sees the next-category or next-region target, so no label can travel from test to train through it. Second, because the graph is built over all places, the only test-side exposure is graph structure. We measured it directly, rebuilding the representation per fold from its training users only and re-running both tasks against the full-corpus version on the same folds. The effect is within fold noise, with no systematic gain: on region the change is -0.33 points at Alabama, $+0.01$ at Arizona, and -0.12 at Florida. On category it is $+0.29$ at Alabama, $+0.27$ at Arizona, and $+0.00$ at Florida, measured on the in-coverage visits whose places appear in training (about 67 percent at Alabama to 87 percent at Florida). This is a place-level proxy and the one residual we cannot fully measure: the representation cannot score a place never seen in training; on the measurable visits, the large majority, the audit finds no inflation. Third, the one component that could pass visit-to-visit information is a region-transition prior, a table of how often one region follows another that biases region scores; an earlier version built on the whole dataset inflated region accuracy by 13 to 27 points across states, so any such table is built per fold, from training data only, seed by seed. The joint and dedicated models do not use this prior; it survives only in the HMT-GRN baseline (Section V-D). The place-level baseline representation (HGI) is pre-trained once on the whole dataset, like ours; the contextual baseline (CTLE) is pre-trained per fold on training users only, so no baseline input is held to a weaker standard than ours.

C. Metrics, superiority versus non-inferiority

For the category task we report macro-averaged F1 (macro-F1), which averages the per-class F1 so each of the seven categories counts equally; its reference point is the majority-class floor, the macro-F1 of always predicting the most common category. For the region task we report accuracy at ten (Acc@10), the fraction of test visits whose true region appears among the model’s ten highest-scoring guesses (a visit whose true region is absent from that fold’s training data counts as an error); its comparison anchor is the dedicated single-task model.

Where the joint model is meant to outperform, we test *superiority*: at the four datasets with four seeds, a paired test over the per-seed means with a Holm correction across the

cells; at the two single-seed states, a paired Wilcoxon signed-rank test over the folds. Where it is meant only to stay even, we test *non-inferiority*, that it is no worse than the dedicated model by more than a two-point margin, with a two-one-sided-tests (TOST) procedure. We fix the two-point margin in advance, on deployment grounds: a mobility-aware service acts on which region will be busy, not on a single rank position (Section III), so a two-point shift in Acc@10 is below the granularity at which it would behave differently (for scale, with five hundred twenty to eight thousand five hundred regions, a random top-ten guess lands the true region at most about two percent of the time). The equivalence is well powered: the paired difference has a small standard deviation (0.04 to 0.15 points at the four multi-seed datasets, 0.07 at the two single-seed ones), giving power near 1.0 to reject a true two-point gap, so a non-inferiority verdict reflects a real match.

D. Baselines

The comparisons play three roles. The first is the per-task state of the art, on our data, under user-disjoint five-fold protocols. For next-category this is a faithful re-implementation of POI-RGNN [29] and a Markov baseline over recent categories, reporting the strongest order per dataset (Markov-K, up to the nine-visit window). For next-region this is HMT-GRN [16], our primary comparison, an end-to-end recurrent model run on the same data, seed, and folds; we keep its shared multi-task skeleton, add a region-transition prior built per fold from training data, and drop its graph components and hierarchical beam search, which serve the next-place head we do not predict. It is a region-native model, not a reproduction of the complete published system. We also run a faithful STAN [12], trained from raw with its own embeddings and sequence construction under one audited recipe, and a ReHDM [30] reference reported under its own published protocol. The second is two representation controls attributing the category gain to our specific design, not to contextualization in general or extra features: CTLE [10], the closest prior contextual check-in embedding, run end to end at Florida and in frozen form at Alabama, Arizona, and Istanbul; and a feature-concatenation control appending raw per-visit features to a standard place embedding under the same model. The third compares the two ways of coupling the tasks. The dominant published coupling is the directed cascade, predicting the category first and conditioning the place prediction on it [21], [23]; CSLSL [22], its current form, reports the chain outperforming a parallel variant sharing one trunk. We test the cascade inside our own model rather than re-implementing those systems: we remove the cross-attention stack where the two streams exchange information and feed the predicted category forward into the region pathway, keeping the representation, outputs, and training recipe unchanged. This isolates the coupling topology (chain versus parallel) at no additional parameter or compute cost: the chain form drops the exchange stack and adds a conditioning projection of about a thousand parameters. The standard place-level embedding [8] isolates what the per-

visit representation adds. The full training recipe for every model ships with the released code (Section VIII).

VI. RESULTS

A. The representation makes the category task learnable

Table II compares our check-in-level representation against a standard place embedding (HGI [8]) on next-category macro-F1. The comparison is like-for-like: target, single-task head, recipe, windowing, and folds are identical; only the input representation changes (a vector per visit versus a vector per place), so the margin isolates the representation. (The check-in-level column therefore keeps one fixed recipe, intentionally not the per-dataset-tuned ceiling of Table III.) The check-in-level representation outperforms the place embedding by a wide margin at every state: +29.31 at Alabama, +27.63 at Arizona, +39.62 at Florida, +37.95 at California, and +37.47 at Texas. The same holds on Istanbul (+28.09). The gains fall in two bands, about +28 to +29 points at the smaller datasets (Istanbul, Alabama, Arizona) and about +37 to +40 at the three large states. A controlled test attributes most of that gain to the per-visit context itself: a per-place-averaged version of the same representation leaves roughly 64 to 90 percent of the gain (state-dependent) to the context each visit carries, not to extra training signal.

The geometry of the vectors shows why the representation helps this task in particular: the per-visit vectors’ silhouette score by category, a -1 to 1 measure of how tight and well separated the seven labeled groups are, is about 0.57, against about 0.00 for the place embedding. Their nearest-neighbor category purity, the share of nearest neighbors carrying the vector’s own category, is about 0.98, against about 0.78 (both averaged over the five U.S. states). The same geometry does not separate regions, so the benefit is category-only (the joint model’s spatial stream instead reads the region-level vectors of the same graph, Section IV-B). CTLE [10], the closest prior contextual embedding, also gives each visit its own vector, but it pretrains on place identifiers and timestamps alone, so the category vocabulary never enters its training, whereas our graph reads each visit’s category and time of day as input features. Fine-tuned end to end at Florida at its authors’ defaults, at the same 64 dimensions and windowing as ours, CTLE reaches 33.45 macro-F1 at its best epoch (29.69 at its final one), about two points below the place embedding under the same best-epoch rule and far below our 75.15; frozen and fed to the same single-task model, it repeats the ordering at Alabama, Arizona, and Istanbul, with margins of +37.8, +37.0, and +28.7 macro-F1. A feature-concat control, the place embedding joined with raw per-visit features and fed to the same model, does not close the gap either, so the category gain comes from the hierarchical per-visit representation, not from contextualization alone or feature injection.

B. One model, two tasks

Read this as: one model outperforms a dedicated next-category model at every dataset and outperforms or matches a dedicated next-region model. To calibrate the two scales:

TABLE I

DATASET STATISTICS, ORDERED BY REGION COUNT: FIVE GOWALLA STATES AND ONE INTERNATIONAL CITY (ISTANBUL, MASSIVE-STEPS). ALL SHARE THE SAME SEVEN ROOT CATEGORIES; THE REGION UNIT IS A CENSUS TRACT (A MAHALLE FOR ISTANBUL). WINDOWS ARE THE OVERLAPPING NINE-VISIT INPUTS PLUS THE NEXT-VISIT TARGET; LENGTHS ARE PER-USER CHECK-IN COUNTS; DENSITY IS THE CHECK-IN-MATRIX FILL, $100 \times \text{CHECK-INS}/(\text{USERS} \times \text{POIS})$; MAJORITY IS THE SHARE OF NEXT-VISIT LABELS IN THE MOST COMMON CATEGORY (FOOD IN EVERY DATASET).

Dataset	Source	Check-ins	Users	POIs	Regions	Windows	Max len	Avg len	Density (%)	Majority (%)
Istanbul	Massive-STEPS	462,615	23,694	29,816	520	271,666	817	19.5	0.065	33.4
AL	Gowalla	113,846	3,858	11,848	1,109	96,326	3,835	29.5	0.249	34.2
AZ	Gowalla	236,450	7,869	20,666	1,547	200,895	5,589	30.1	0.145	34.0
FL	Gowalla	1,407,034	21,052	76,544	4,703	1,274,418	16,679	66.8	0.087	24.7
TX	Gowalla	4,089,892	38,644	160,938	6,553	3,830,414	42,300	105.8	0.066	31.0
CA	Gowalla	3,171,380	37,090	169,145	8,501	2,925,466	14,855	85.5	0.051	32.7

category macro-F1 spans seven classes (a majority-class floor of about 7%); region Acc@10 spans hundreds to thousands of regions, where a random top-ten guess is right at most about two percent of the time (well under one percent at the larger region counts). Table III reports the single joint model against the dedicated single-task ceilings, so called because a jointly trained head is usually expected at best to match its dedicated counterpart. Each dedicated category ceiling is tuned per dataset over batch size and learning rate (the region ceilings use the strongest fixed recipe). Figure 3 plots the signed gains ordered by region count. On category the joint model outperforms the dedicated ceiling at every dataset, by +5.34 to +9.40 macro-F1 (smallest at Florida, largest at Arizona, and +8.59 at Istanbul). On region (Acc@10) the joint model outperforms the dedicated ceiling at Florida +0.72 (4,703 regions), Texas +2.12 (6,553), California +2.20 (8,501), and Istanbul +0.28 (520), and stays a non-inferior match (TOST, ± 2 pp) at Alabama (−0.31) and Arizona (+0.10). Across the five U.S. states the region gain rises with the number of regions, from −0.31 at the smallest to +2.20 at the largest. Region count and corpus size co-vary across these datasets, so we read the trend across the points rather than as a precise law.

Four of the six datasets are measured at $n=20$ on both arms: Alabama, Arizona, Florida, and Istanbul average four seeds over the same five frozen folds, and the tests pair the per-seed means (four paired observations per dataset). On category, all four seeds favor the joint model at each of these datasets, and each per-dataset gain is significant after a Holm correction across the four cells (paired t , corrected $p < 0.001$ everywhere). On region, all four are tested equivalences (TOST, ± 2 pp), and all four pass with 90% confidence intervals well inside two points: Alabama (−0.31; −0.48 to −0.14), Arizona (+0.10; +0.00 to +0.21), Florida (+0.72; +0.67 to +0.78), and Istanbul (+0.28; +0.24 to +0.32). At Florida and Istanbul the whole interval sits above zero, so the joint model outperforms there as well, though both gains are smaller than the two-point margin of Section V-C; the region gains above that margin are at Texas and California. At Arizona the interval touches zero, so we report a match, not a gain. At Alabama the interval sits entirely below zero, a small but statistically significant deficit, still well within the two-point margin. California and Texas remain provisional: their dedicated ceilings are $n=20$, but the joint cells are a

single seed over five folds. On those folds, all five favor the joint model on both tasks (one-sided paired Wilcoxon, $p=0.031$, the smallest value five folds can produce), and the region gains carry 90% intervals entirely above the two-point margin (+2.14 to +2.28 at California, +2.04 to +2.18 at Texas). A multi-seed run at these two states is the remaining confirmation.

A reader used to multi-task learning [6] expects the harder task to pay for the easier one. As a control, we freeze the region pathway at the start of training so it can neither learn nor teach the category task, yet the full category lift survives at Alabama, Arizona, and Florida (within 0.3 of the joint model and far above the dedicated single-task model). We therefore read the category gain as a stronger shared trunk, obtained without a second model to serve (one model, one forward pass), not the region task teaching the category one; we report this as a finding, not a hypothesis.

The joint model is also above every external baseline reported, on both tasks, at every dataset (Table III): the primary region-native comparison, HMT-GRN [16] (for example California 49.61 versus our 65.69), a from-raw STAN [12] and a ReHDM reference [30] on region, and POI-RGNN [29] and a Markov floor on category. These externals run on their own embeddings, so this comparison folds in the representation advantage of Section VI-A; the like-for-like anchor remains the dedicated column. A simple Markov region floor reaches 43 to 65 Acc@10 across the datasets (computed under a non-overlapping windowing of the same data, indicative rather than protocol-matched); the joint model clears it by 12 to 23 points everywhere.

On principle we prefer coupling the tasks in parallel rather than in a chain: the two targets are co-equal, so a chain must pick one to predict first and pass its errors to the other. The choice still had to be checked, because CSLSL reports its chain outperforming a shared-trunk parallel variant on its own benchmarks [22]. Rewired as a cascade (Section V-D), our model reaches the same combined score, the geometric mean of the two task metrics, as its parallel form at no additional parameter or compute cost: at Alabama, Arizona, Florida, and Istanbul, with both forms trained under the same configuration and single seed, the difference is about zero and neither task moves by more than a quarter of a point. We read this as a defense of the parallel design, not a claim that we outperform the cascade: on this representation the coupling

TABLE II

NEXT-CATEGORY MACRO-F1 OF THE CHECK-IN-LEVEL REPRESENTATION AGAINST A STANDARD PLACE EMBEDDING (HGI), SAME SINGLE-TASK HEAD, RECIPE, AND FOLDS (MEAN AND FOLD STANDARD DEVIATION, SEED 0).

Dataset	Check-in level	Place level	Δ
Istanbul	54.65 ± 0.6	26.56 ± 0.5	+28.09
AL	55.87 ± 2.4	26.56 ± 1.0	+29.31
AZ	57.13 ± 1.3	29.50 ± 1.0	+27.63
FL	75.15 ± 0.5	35.53 ± 0.3	+39.62
TX	69.95 ± 0.2	32.48 ± 0.6	+37.47
CA	70.26 ± 0.3	32.31 ± 0.4	+37.95

The matching place-level value at Alabama and Istanbul (26.56) is a coincidence of two independent runs; their per-fold values differ.

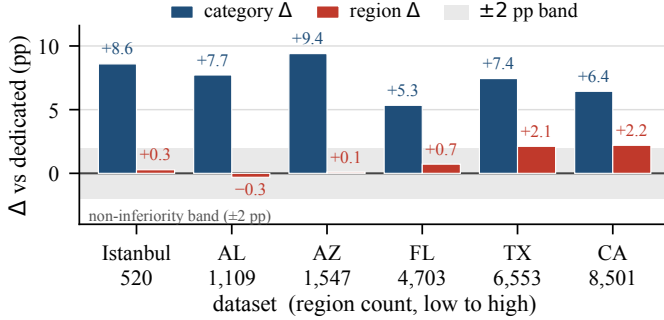


Fig. 3. Signed gains of the joint model over the dedicated single-task models, ordered by region count. The category gain (macro-F1) is positive at every dataset; the region gain (Acc@10) rises with region count across the five U.S. states and is also positive at Istanbul. The two metrics are on different scales; the ± 2 -point band applies to region.

topology makes no measurable difference, and the simpler parallel structure loses nothing. Two qualifications bound this reading: the cascade runs under the recipe tuned for the parallel model, and its form was fixed in advance, so a carefully tuned cascade remains future work.

C. External validity (Istanbul)

Read this as: the finding holds on a non-U.S. city under the same representation, overlapping-window protocol, and training setup as the U.S. states. On Istanbul the joint model outperforms the dedicated category ceiling by +8.59 macro-F1 (54.74 to 63.33) and edges past the dedicated region ceiling at +0.28 Acc@10 (75.16 to 75.44), a statistically supported gain (the 90% confidence interval of the paired difference lies entirely above zero; it also passes the non-inferiority test, TOST, ± 2 pp); both arms average four seeds ($n=20$). The comparable quantity is the lift over the dedicated ceiling, not the absolute Acc@10, since region counts differ across datasets (520 mahalle here). The external baselines line up the same way: on category the joint model is far above the Markov floor (24.55) and POI-RGNN [29] (30.12), and on region it is above ReHDM [30] (69.33), a faithful STAN [12] (61.86), and HMT-GRN [16] (60.4). The U.S. picture repeats on a different continent and region unit.

VII. DISCUSSION AND LIMITATIONS

One model is enough. A shared trunk carries the semantic context that lifts the next-category task sharply over a dedicated model at every dataset, and a private spatial path keeps the next-region task competitive: a non-inferior match (TOST, ± 2 pp) at the two small U.S. states, and a tested gain at the other four datasets. Across the U.S. states the region gain rises with the number of regions (Figure 3). The joint model matches or outperforms the dedicated single-task model on both tasks at once (Table III): one model, one forward pass, anticipates both what a user will do next and where.

As a usage sketch, a service could treat the model’s top-ranked next regions as an anticipatory set: at California, a shortlist of ten regions out of 8,501 contains the true next region 65.69 percent of the time, more than five hundred times better than picking ten at random; at Alabama, ten out of 1,109 capture it 69.80 percent of the time. On the four datasets we measured (Alabama, Arizona, Florida, and Istanbul; a single seed over five folds), the ten shortlisted regions sit about 3 to 8 kilometers from the shortlist’s own center (median across visits, from region centroids). When the top guess is wrong and the true region was seen in training, the predicted region’s centroid sits a similar median of about 3 to 8 kilometers from the true region’s, (visits to regions never seen in training are rarer and miss by more), while two randomly drawn regions from the same map sit a median of 20 to 241 kilometers apart. This remains motivation, not a measured service result.

Three limits qualify these results. First, the joint numbers at California and Texas come from a single seed over five folds, so those two cells are provisional; the other four datasets average four seeds on both arms. Second, our representation is trained once over all places; visits to places never seen during training are the single effect we cannot fully isolate. A planned follow-up trains it on each fold’s training places only and embeds unseen visits directly. Third, although a mobility-aware service motivates this work, we do not build or evaluate one.

Finally, our representation is a fixed per-visit vector that any downstream model can consume. Structural analysis of check-in networks has itself named machine learning as a next step [5]; learning over check-in structure, as this study does, moves in that direction.

VIII. CONCLUSION

We asked whether one model can predict both the category and region of a user’s next visit. First, a check-in-level representation, one vector per visit rather than one per place, makes the next-category task far more learnable than the standard place-level representation does. Second, on that representation, a single multi-task model outperforms a dedicated category model at every dataset and a dedicated region model at four of the six datasets, remaining non-inferior (TOST, ± 2 pp) at the other two. One model with a shared semantic context and a private spatial path anticipates both what a user will do next and where; across the U.S. states, the region gain grows with the number of regions. The code for the model, representation,

TABLE III

ONE MODEL, TWO TASKS: THE SINGLE JOINT MODEL AGAINST THE DEDICATED SINGLE-TASK MODELS AND THE EXTERNAL BASELINES, ORDERED BY REGION COUNT. CATEGORY IS MACRO-F1, REGION IS ACC@10. BOLD MARKS A STATISTICALLY SUPPORTED IMPROVEMENT OVER THE DEDICATED MODEL; IN THE REGION COLUMNS, † MARKS THAT SUPPORTED IMPROVEMENT AND ≈ A NON-INFERIOR MATCH (TOST, ± 2 PP; SECTION VI-B).

Dataset	Regions	Next-category (macro-F1)				Next-region (Acc@10)				
		Markov-K	POI-RGNN	Dedicated	Joint (ours)	HMT-GRN	ReHDM	STAN	Dedicated	Joint (ours)
Istanbul	520	24.55	30.12	54.74 $\pm$ 0.09	63.33 $\pm$ 0.02	60.4	69.33	61.86	75.16 $\pm$ 0.01	75.44 $\pm$ 0.04 [†]
AL	1,109	20.50	23.80	56.82 $\pm$ 0.03	64.54 $\pm$ 0.10	57.05	65.38	60.72	70.11 $\pm$ 0.10	69.80 $\pm$ 0.05 [≈]
AZ	1,547	23.92	27.64	56.43 $\pm$ 0.10	65.84 $\pm$ 0.02	43.70	53.00	49.86	59.46 $\pm$ 0.08	59.56 $\pm$ 0.05 [≈]
FL	4,703	29.74	34.49	74.51 $\pm$ 0.03	79.85 $\pm$ 0.03	63.74	64.49	72.99	76.70 $\pm$ 0.02	77.42 $\pm$ 0.03 [†]
TX	6,553	28.67	33.03	69.79 $\pm$ 0.08	77.23 [°] $\pm$ 0.12	53.85	48.81 [‡]	61.67 [†]	64.95 $\pm$ 0.01	67.07 [°] $\pm$ 0.45 [†]
CA	8,501	27.58	31.78	70.60 $\pm$ 0.07	77.04 [°] $\pm$ 0.20	49.61	50.26 [‡]	58.52 [†]	63.49 $\pm$ 0.02	65.69 [°] $\pm$ 0.30 [†]

HMT-GRN (region-native, on our folds, scored on the visits whose region appears in training) is the primary external comparison; STAN (faithful, from raw) and ReHDM (its own protocol) are references. Markov-K reports the strongest order per dataset (up to the nine-visit window). †STAN partial-fold cells, truncated by training cost: Texas four of five folds, California two of five (seed 0). ‡ReHDM at Texas and California is a single seed. °The joint cells at California and Texas are a single seed over five folds (provisional); every other cell in our columns averages four seeds. $\pm$ is the sd across seeds for four-seed cells, across folds for single-seed cells.

baselines, and statistical tests is available at <https://github.com/VitorHugoOli/PoiMtlNet/tree/mobiwac>; both data sources are already public: the category-annotated Gowalla dump (https://figshare.com/articles/dataset/gowalla_data/22126586) and the Massive-STEPS collection [28].

REFERENCES

- [1] E. Cho, S. A. Myers, and J. Leskovec, "Friendship and mobility: User movement in location-based social networks," in *Proc. ACM SIGKDD (KDD)*. ACM, 2011, pp. 1082–1090.
- [2] C. Song, Z. Qu, N. Blumm, and A.-L. Barabási, "Limits of predictability in human mobility," *Science*, vol. 327, no. 5968, pp. 1018–1021, 2010.
- [3] E. Bastug, M. Bennis, and M. Debbah, "Living on the edge: The role of proactive caching in 5G wireless networks," *IEEE Commun. Mag.*, vol. 52, no. 8, pp. 82–89, 2014.
- [4] C. L. Vielhaus, J. V. S. Busch, P. Geuer, A. Palaos, J. Rischke, D. F. Külzer, V. Latzko, and F. H. P. Fitzek, "Handover predictions as an enabler for anticipatory service adaptations in next-generation cellular networks," in *Proc. ACM MobiWac*. ACM, 2022, pp. 19–27.
- [5] D. L. L. Moura, A. L. L. Aquino, and A. A. F. Loureiro, "On the design of mobility-aware systems: A tourist's perspective," in *Proc. IEEE/ACM MSWiM*, 2025, pp. 667–674.
- [6] R. Caruana, "Multitask learning," *Machine Learning*, vol. 28, no. 1, pp. 41–75, 1997.
- [7] V. H. O. Silva, I. F. Almeida, T. S. Paiva, G. B. Santos, F. A. Silva, and F. T. Sousa, "An investigation into multi-task learning for point-of-interest category classification and next-POI prediction," in *Anais do CBIC*, 2025.
- [8] W. Huang, D. Zhang, G. Mai, X. Guo, and L. Cui, "Learning urban region representations with POIs and hierarchical graph infomax," *ISPRS J. Photogramm. Remote Sens.*, vol. 196, pp. 134–145, 2023.
- [9] P. Veličković, W. Fedus, W. L. Hamilton, P. Liò, Y. Bengio, and R. D. Hjelm, "Deep graph infomax," in *Proc. ICLR*, 2019.
- [10] Y. Lin, H. Wan, S. Guo, and Y. Lin, "Pre-training context and time aware location embeddings from spatial-temporal trajectories for user next location prediction," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, 2021, pp. 4241–4248.
- [11] J. Feng, Y. Li, C. Zhang, F. Sun, F. Meng, A. Guo, and D. Jin, "Deep-Move: Predicting human mobility with attentional recurrent networks," in *Proc. Web Conf. (WWW)*, 2018, pp. 1459–1468.
- [12] Y. Luo, Q. Liu, and Z. Liu, "STAN: Spatio-temporal attention network for next location recommendation," in *Proc. Web Conf. (WWW)*, 2021, pp. 2177–2185.
- [13] S. Yang, J. Liu, and K. Zhao, "GETNext: Trajectory flow map enhanced transformer for next POI recommendation," in *Proc. ACM SIGIR*, 2022, pp. 1144–1153.
- [14] M. Luca, G. Barlacchi, B. Lepri, and L. Pappalardo, "A survey on deep learning for human mobility," *ACM Comput. Surv.*, vol. 55, no. 1, pp. 7:1–7:44, 2021.
- [15] Z. Sun, Y. Lei, L. Zhang, C. Li, Y.-S. Ong, and J. Zhang, "A multi-channel next POI recommendation framework with multi-granularity check-in signals," *ACM Trans. Inf. Syst.*, vol. 42, no. 1, pp. 1–28, 2024.
- [16] N. Lim, B. Hooi, S. Ng, Y. L. Goh, R. Weng, and R. Tan, "Hierarchical multi-task graph recurrent network for next POI recommendation," in *Proc. ACM SIGIR*, 2022, pp. 1133–1143.
- [17] N. Zhu, J. Cao, X. Lu, C. Liu, H. Liu, Y. Li, X. Luo, and H. Xiong, "Predicting a person's next activity region with a dynamic region-relation-aware graph neural network," *ACM Trans. Knowl. Discov. Data*, vol. 16, no. 6, pp. 1–23, 2022.
- [18] K. Sun and M. Xu, "Knowledge graph tokenization for behavior-aware generative next POI recommendation," arXiv:2509.12350, 2025, arXiv preprint.
- [19] D. Liao, W. Liu, Y. Zhong, J. Li, and G. Wang, "Predicting activity and location with multi-task context aware recurrent neural network," in *Proc. IJCAI*, 2018, pp. 3435–3441.
- [20] Y. Wang, H. Chen, S. Liu, J. Zhang, L. Wu, X. Cui, and Y. Hu, "Hierarchy aware-based multi-task learning for user location prediction," *The Journal of Supercomputing*, vol. 81, no. 11, 2025.
- [21] J. Ye, Z. Zhu, and H. Cheng, "What's your next move: User activity prediction in location-based social networks," in *Proc. SIAM SDM*, 2013, pp. 171–179.
- [22] Z. Huang, S. Xu, M. Wang, H. Wu, Y. Xu, and Y. Jin, "Human mobility prediction with causal and spatial-constrained multi-task network," *EPJ Data Science*, vol. 13, no. 1, p. 22, 2024.
- [23] F. Yu, L. Cui, W. Guo, X. Lu, Q. Li, and H. Lu, "A category-aware deep model for successive POI recommendation on sparse check-in data," in *Proc. Web Conf. (WWW)*, 2020, pp. 1264–1274.
- [24] T. Yu, S. Kumar, A. Gupta, S. Levine, K. Hausman, and C. Finn, "Gradient surgery for multi-task learning," in *Proc. NeurIPS*, vol. 33, 2020.
- [25] A. Navon, A. Shamsian, I. Achituve, H. Maron, K. Kawaguchi, G. Chechik, and E. Fetaya, "Multi-task learning as a bargaining game," in *Proc. ICML*, 2022, pp. 16428–16446.
- [26] D. Xin, B. Ghorbani, A. Garg, O. Firat, and J. Gilmer, "Do current multi-task optimization methods in deep learning even help?" in *Proc. NeurIPS*, vol. 35, 2022.
- [27] T. H. Silva, A. C. Viana, F. Benevenuto, L. Villas, J. Salles, A. A. F. Loureiro, and D. Quercia, "Urban computing leveraging location-based social network data: A survey," *ACM Comput. Surv.*, vol. 52, no. 1, pp. 17:1–17:39, 2019.
- [28] W. Wongso, H. Xue, and F. D. Salim, "Massive-steps: Massive semantic trajectories for understanding POI check-ins — dataset and benchmarks," arXiv:2505.11239, 2025.
- [29] C. G. S. Capanema, G. S. de Oliveira, F. A. Silva, T. R. de Moura Braga Silva, and A. A. F. Loureiro, "Combining recurrent and graph neural networks to predict the next place's category," *Ad Hoc Networks*, vol. 138, p. 103016, 2023.
- [30] X. Li, Z. Gu, R. Yao, Y. Zhou, H. Zhu, J. Zhao, and W. Du, "Beyond individual and point: Next POI recommendation via region-aware dynamic hypergraph with dual-level modeling," in *Proc. IJCAI*, 2025, pp. 3081–3089.